

Toxicity Risk -> Assay Recommender

A retrieval-based tool that scores a small-molecule candidate's toxicity risk by its similarity to drugs that failed in the clinic (or were withdrawn) for toxicity, and recommends which in-vitro assays to run first.

Positioning in one line: *"We retrieve the clinically-failed drugs your candidate most resembles — including non-obvious analogs — show the shared structural motif, and rank which in-vitro assays would catch the implicated liability first. It prioritizes experiments; it does not declare a molecule safe."*

0. Project context & status (read first)

- **Event:** a bio × AI hackathon. Deliverable is an **MVP + investor/judge-facing demo** (see §6), not a validated product.
 - **How we got here:** this idea was chosen over an alternative — a *cross-specificity / immunogenicity risk tool for biologics* (antibody candidates, population-specific pre-sensitization, e.g. cetuximab/α-gal) — because the small-molecule tox tool is **more tractable**: all core datasets are downloadable CSVs and the MVP is retrieval + light reasoning with no model training required. The biologics idea is a possible fallback, not the current plan.
 - **Repo state: greenfield.** As of writing, /Users/cheparukhin/hackathon/ contains only this doc — no code yet. Stack/infra not yet chosen.
 - **Guiding principle:** ship the **floor** (§4 Layer 1 ECFP4 baseline) first so there is always a working demo; treat learned embeddings, 3D, and AACT mining as differentiators/stretch.
 - **Parameters still to fill (TBD — not yet decided in-conversation):**
 - Timeframe / submission deadline.
 - Team size and strengths (eng vs. cheminformatics vs. bio) — affects how far up the stretch layers to reach.
 - Explicit judging criteria (working product vs. scientific novelty vs. commercial potential).
 - Target LLM + vector store + hosting for the demo.
 - **Data caveat:** dataset sizes below are **last-published figures** (some flagged “verify”); several sources grow continuously or come from secondary summaries. **Re-confirm exact counts against the primary source before quoting them in a pitch** (especially ClinTox positive-class count, DrugBank withdrawn count, ToxCast endpoint total).
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1. The idea

- **Input:** a small-molecule drug candidate (SMILES; optionally 3D conformer).
- **Reference database:** molecules with known organ-specific toxicity, weighted toward those that **failed clinical trials or were withdrawn for toxicity** (not just assay-positive compounds).
- **Output:**
 1. A **per-modality risk vector** — {liver: 0.71, cardio: 0.20, hERG: 0.10, kidney: 0.05, ...}.
 2. A **ranked in-vitro assay panel** — e.g. *“prioritize hERG + iPSC-cardiomyocyte (driven by 0.71 similarity to [drug X, DICTrank-Most-concern]); then hepatocyte spheroid (0.58).”*
 3. An **evidence trail** — the nearest failed drugs, the organ they failed in, and the **shared structural motif / toxicophore** driving the match.

It is a **decision-support / experiment-prioritization** tool, not a binary safety classifier. That framing is both more scientifically honest (structure alone misses metabolism, off-target, and dose-driven toxicity) and easier to validate and adopt.

2. Why it's novel (white space + prior art to differentiate from)

The two halves of the idea each exist separately; the **combination is unoccupied**:

- **Endpoint predictors** — ProTox 3.0, admetSAR 3.0 / ADMETlab 3.0, ADMET-AI, DeepTox, DEREK. They predict *toxicity probabilities / structural alerts*, **not “which assay to run,”** and are trained on heterogeneous tox data, **not grounded in clinical-trial failures**.
- **Read-across engines** — EPA **GenRA**, OECD QSAR Toolbox, AMBIT. Analog-based tox inference (methodologically the closest cousin), but reference sets are **environmental/regulatory chemicals, not clinically-failed drugs**, and output isn’t a prioritized clinical assay panel.
- **Closest single paper** — “*Comprehensive Analysis of Clinically Discontinued Compounds Using an In Vitro Secondary Pharmacology Panel*” (PMC12441832, 2024): took **52 small molecules discontinued in Phase 1–3 for safety** and ran them through a **47-target / 78-assay** panel. Crucially it screens **everything** and argues the primary target does *not* predict off-target risk — it does **not** use structural similarity to *prioritize a subset* of assays. **That is exactly our niche.**

A combined literature query {chemical similarity + clinical-trial failure + toxicity + recommend + in-vitro assay + read-across} returned **0 results** — suggestive that the exact framing is unpublished.

Our two defensible novelty claims: 1. **Reference set grounded in *clinical* toxicity failures** (not assay-positives). ClinTox already operationalizes this; WITHDRAWN + Probes&Drugs strengthen it. 2. **Output = a *specific, ranked assay panel***, made credible by the organ-resolved label layer (DILrank->liver assays, DICTrank/hERG->cardiac assays, Tox21/ToxCast pathways->named HTS assays).

The genuine white space is the **linkage layer**: candidate -> nearest clinically-failed neighbors -> the organ/mechanism they implicated -> the specific assay(s) that would have caught them.

3. Datasets

3a. Clinical-failure reference set (the novelty; assemble, don’t download whole)

No single public dataset cleanly says “these N molecules died in the clinic *because of toxicity*, here are their SMILES.” We build it by merging:

Dataset	Content	Size	Access	Role
ClinTox (MoleculeNet / TDC)	FDA-approved vs. failed clinical trials for toxicity	~1,491 cpds w/ SMILES; positives (tox-failures) only ~90–110	Free — <code>deepchem.molnet.lodopositives.tdc</code> , TDC, HF <code>zpn/clintox</code>	Start here. Its positives were mined from AACT — i.e. our exact idea, already v0. Class-imbalanced.
WITHDRAWN 2.0 (Charité)	Withdrawn/discontinued drugs, withdrawal reasons classified by organ toxicity	578 (v1); 2.0 adds FAERS + mechanism	Free — <code>chem-info.charite.de/withdrawn</code> , NAR 2024 52:D1503	Best organ-resolved failure set. Higher label quality than ClinTox.
Probes & Drugs — withdrawn set	Withdrawn drugs pre-joined to ChEMBL/DrugBank structures	~633 (v02.2025)	Free — <code>probes-drugs.org/compounds</code>	De-dupe / canonical-SMILES join across sources.
DrugBank (withdrawn flag)	withdrawn category, full structures	~250–300 (verify)	Free academic; non-commercial license	Structure + flag; weaker on “reason = tox”.

Dataset	Content	Size	Access	Role
AACT (ClinicalTrials.gov dump)	why_stopped free-text, TERMI-NATED/SUSPENDED, interventions	full CT.gov	Free — aact.ctti-clinicaltrials.org	Stretch goal. NLP-classify tox terminations -> resolve drug names to SMILES (~30–60% resolvable). Demo on a slice; don't rebuild wholesale.

3b. Organ-specific toxicity labels (power the “which assay” recommendation)

FDA-curated *ranked* lists — the highest-quality organ-resolved labels; the backbone of the per-modality score.

Dataset	Content	Size	Assay it points to
DILrank (FDA LTKB)	Drug-induced liver injury , 4 concern classes	~1,036 (v1); DILrank 2.0 = 1,336	Hepatocyte / spheroid, BSEP inhibition, mito-tox same
DIList (FDA)	Binary hepatotoxic (broader)	1,279	
DICTrank (FDA, 2023)	Drug-induced cardiotoxicity , 4 concern levels	1,318 (1,291 approved + 27 withdrawn)	hERG, iPSC-cardiomyocyte, Nav1.5/Cav1.2
hERG datasets	hERG channel block	TDC 648; DeepHIT ~14k; UnihERG ~20k	hERG patch-clamp / automated electrophysiology
SIDER	Side effects mined from labels	1,556 drugs, 5,868 SE terms	phenotype -> organ -> assay
OFFSIDES / TWOSIDES	FAERS-derived off-label / DDI side effects	438,801 / 868k assocs	pharmacovigilance signal layer
LiverTox (NIH)	Narrative hepatotoxicity monographs	~1,200 agents	qualitative liver context (text)

(!) SIDER/OFFSIDES/FAERS/DILrank describe drugs that **reached market** (survivorship bias). Use them as **organ-tox labels**, not as the “clinical-failure” reference set.

3c. Assay / mechanism datasets (map similarity -> a named assay)

Dataset	Content	Size	Role
Tox21	12 pathway assays (nuclear-receptor + stress)	~7,800 cpds × 12	Each pathway is an assay -> direct “which assay” signal
ToxCast (EPA)	High-throughput in-vitro endpoints	~800+ endpoints × >4,200 chem	Richest public compound -> assay-endpoint matrix (mechanistic layer). Mostly environmental chemicals — partial drug overlap
ChEMBL	Measured bioactivities incl. safety targets	~2.4M cpds	Off-target / safety-pharmacology activities; MCP available for live pulls

Dataset	Content	Size	Role
(from prior notes) TG-GATES / Open TG-GATES / DrugMatrix / LINCS L1000	Toxicogenomics / histopath / cell response	~170 / ~600 / >20k	Mechanism enrichment (stretch)

3d. Structure sources (SMILES / fingerprints / name-resolution)

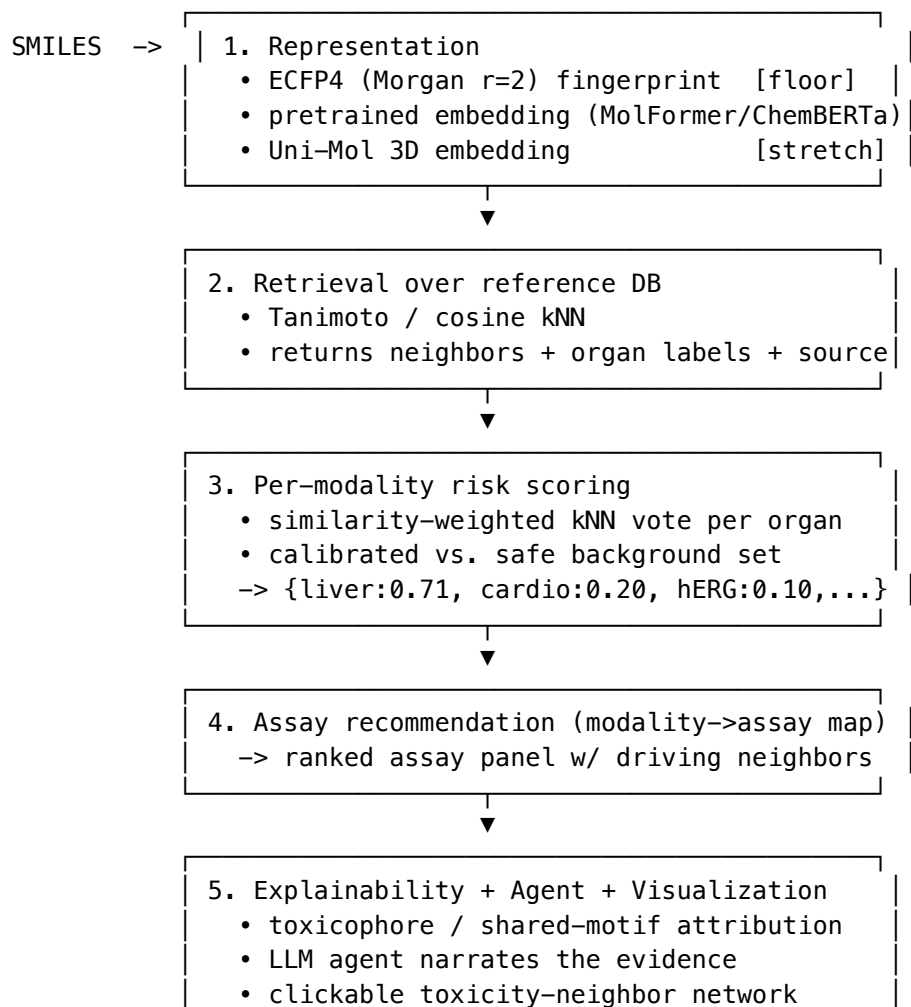
- **PubChem** (public domain, >100M) — best for resolving trial/FDA-list **drug names -> structure**.
- **ChEMBL** (CC-BY-SA) — structures + bioactivity; **MCP available**.
- **DrugBank** (free academic, non-commercial) — best annotations.

3e. Background / negative set (do not skip)

Approved, chronically-used “safe” drugs (the **ClinTox approved class** works). Required so the score means “*enriched for toxic neighbors relative to safe neighbors*,” not merely “*similar to some molecule*.” Without a background, every candidate looks risky because everything has a nearest neighbor.

4. Architecture

Layered so there is always a working floor even if the higher layers slip.



Reference DB schema (organ/mechanism-resolved, multi-modal)

compound_id | canonical_SMILES | source(s) | fail_stage | clinical_failure_flag |
liver_risk | cardio_risk | hERG | kidney | neuro | hematologic | ... |
mechanism (optional) | representative_evidence_link

- Store labels as a **vector across modalities**, not a single “reason” string — a molecule can carry several liabilities.
- `clinical_failure_flag` (from ClinTox/WITHDRAWN) is a separate boost/badge so we keep the clinical-failure story *and* the score stays credible on the cleaner FDA organ ranks.

Layer 1 — Representation (the “non-obvious similarity” differentiator, honestly scoped)

- **Floor:** ECFP4 + Tanimoto. Guaranteed to run; explainable; ships.
- **Differentiator:** add **one** pretrained encoder — **MolFormer or ChemBERTa** (SMILES, via HuggingFace, no conformers = easiest). Show a case where the learned embedding and fingerprints **disagree** -> the “finds what chemists miss” moment.
- **Honesty caveat:** an off-the-shelf pretrained embedding is a *general-chemistry* space, **not a “toxicity space.”** True toxicity clustering requires supervised/contrastive fine-tuning on the labels (>1-weekend, gated by ~1–3k labeled cpds). Do **not** over-claim “learned toxicity space” for the MVP.
- **Stretch:** Uni-Mol 3D — only if someone owns conformer generation (3D similarity is conformer-dependent and a time sink).

Layer 3 — Scoring

Per modality m : $\text{score}_m = \sum_i \text{sim}(\text{candidate}, \text{neighbor}_i) \cdot \text{label}_{\{i,m\}}$ over top-k neighbors, normalized against the safe-background neighbor distribution (enrichment, not raw similarity).

Layer 4 — Modality -> assay mapping (hand-authored; the payoff)

Modality flagged	Recommended assays (priority order)
Liver (DILrank)	Human hepatocyte / 3D spheroid -> BSEP (bile salt export pump) inhibition -> mitochondrial toxicity (Glu/Gal, Seahorse)
Cardiac (DICTrank)	hERG patch-clamp -> iPSC-cardiomyocyte (contractility / field potential) -> Nav1.5 / Cav1.2
Kidney	Proximal tubule cells / kidney organoid -> transporter (OAT/OCT) assays
Reactive metabolite	Glutathione-trapping / covalent-binding assay; CYP-dependent cytotox
Tox21/ToxCast pathway hit	the specific named HTS assay for that pathway (e.g. mitochondrial membrane potential, ARE, ER/AR)
Hematologic / neuro / lung	organ-appropriate primary/iPSC model (from neighbor labels)

Layer 5 — Explainability, agent, visualization

- **Toxicophore attribution** — return the substructure / motif driving each match (“*shares an aromatic sulfonamide + exposed tertiary amine with cholestatic-injury compounds*”). Methods: fingerprint-bit -> atom mapping (trivial for ECFP), or attention / integrated gradients / GNNExplainer for the learned model. **Highest-leverage single feature** — it’s the defense against “isn’t this just fingerprint similarity?”
- **LLM agent** interprets retrieved evidence into a report (retrieval + reasoning; **no model training needed for v1**). Optional multi-role framing for the pitch: Retriever -> Evidence -> Hypothesis -> Experiment-planner.

- **Visualization** — candidate at center, historical neighbors around it, organ-colored (*liver* kidney *heart); click a node -> why it failed, shared motif, literature, suggested assay.
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5. Roadmap (hackathon-scoped, layered by risk)

Phase 0 — Data assembly (do first; this is the real work) - [] Ingest ClinTox, WITHDRAWN 2.0, Probes&Drugs withdrawn, DILIrank 2.0, DICTrank, a hERG set. - [] Resolve names -> canonical SMILES via PubChem/ChEMBL; de-dupe. - [] Build the modality-labeled reference table + the safe background set (ClinTox approved). - [] **Curate ≥ 1 verifiable “non-obvious analog -> real clinical tox failure” pair** for the demo (see Risks — this is the single most important demo asset; a wrong neighbor when a judge clicks kills the pitch).

Phase 1 — Floor (must ship) - [] ECFP4 + Tanimoto kNN retrieval. - [] Per-modality risk vector, background-calibrated. - [] Modality->assay mapping -> ranked assay panel + evidence trail.

Phase 2 — Differentiator - [] Add MolFormer/ChemBERTa embedding + retrieval; surface a fingerprint-vs-learned disagreement. - [] Toxicophore / shared-motif highlighting. - [] LLM agent report + clickable neighbor network.

Phase 3 — Stretch (only if time) - [] AACT why_stopped tox-termination mining on a slice (“how the reference set scales”). - [] Uni-Mol 3D embedding. - [] Cross-check overlay vs. ProTox / ADMET-AI as an independent second opinion.

6. Demo / pitch (investor-facing)

Don’t claim “our AI predicts toxicity better than pharma” (needs years of validation). Claim: **“our AI reasons like an experienced med chemist — finds non-obvious precedents, explains them, and recommends the next experiment.”** Demo flow:

1. Paste a candidate (e.g. a kinase inhibitor).
2. *Analyzing structure... searching N clinically-failed compounds... found non-obvious cluster.*
3. Reasoning: *“shares a 3D arrangement of aromatic rings + tertiary amine with cholestatic-injury compounds; the scaffold differs, so fingerprint search would miss this.”*
4. Ranked assay panel (hepatocytes -> BSEP -> iPSC-cardiomyocyte).
5. Organ-colored neighbor network; click to inspect a failed drug.

Value prop: broad safety panels are expensive — “move renal assays to the front of the queue” saves cost, time, animal studies, and catches liabilities earlier. Easier to validate than binary safety claims because it **prioritizes tests**.

7. Risks & honest framing (raise these before a judge does)

1. **Small positive class.** ClinTox has only ~90–110 tox-failures; lean on the FDA organ ranks (DILI-rank/DICTrank, ~1.3k each) as the scoring backbone and use clinical-failure flags as a boost.
 2. **“Failed in trials” label is noisy** and not always organ-resolved. WITHDRAWN’s organ tags mitigate.
 3. **Structure alone misses metabolism / reactive metabolites / off-target / dose-driven tox.** -> position as *prioritization*, never a safety verdict.
 4. **Off-the-shelf embedding $\neq$ “toxicity space.”** Don’t over-claim without supervised fine-tuning.
 5. **The demo needs a real non-obvious analog.** Fabricated clusters collapse under a click-through. Curating a true one is Phase-0 work.
 6. **Proprietary gap.** The cleanest “discontinued-for-safety” attrition data (Pharmaprojects/Citeline, Pharmapendium) is paywalled — the 52-compound paper used it. Note as “gold standard if licensed,” out of hackathon scope.
 7. **DrugBank / SIDER / OFFSIDES licenses are non-commercial** — flag if the output could be commercialized.
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8. Tools

RDKit (fingerprints + Tanimoto + substructure), HuggingFace (MolFormer/ChemBERTa), Uni-Mol (3D, stretch), a vector store (FAISS or in-memory for ~5k cpds), an LLM for the agent report. Cross-check baselines: ADMET-AI, ProTox 3.0, admetSAR/ADMETlab 3.0, EPA GenRA.

9. Key sources

- ClinTox — MoleculeNet, arXiv 1703.00564; TDC single_pred/Tox
- WITHDRAWN 2.0 — NAR 2024;52:D1503 (v1: NAR 2016;44:D1080)
- Probes & Drugs — probes-drugs.org/compoundsets
- AACT — aact.ctti-clinicaltrials.org; terminated-trials analysis PMC4444136
- DILIrank 2.0 / DILIST — FDA LTKB; DILIST ScienceDirect S1359644619303824
- DICTrank — FDA (2023); Drug Discov Today S1359644623002866
- Tox21 / ToxCast — tox21.gov; EPA figshare invitrodb
- SIDER / OFFSIDES / TWOSIDES — nsides.io; Tatonetti Lab
- Closest prior art — secondary-pharmacology of discontinued compounds, PMC12441832 (2024)
- Read-across — EPA GenRA
- Predictors — ADMET-AI (admet.ai.greenstonebio.com); ProTox 3.0 NAR 2024 W513; admetSAR 3.0 / AD-METlab 3.0 NAR 2024 W422
- Encoders — Uni-Mol; MolFormer; ChemBERTa